Tight binding model assignment

1. Nearest neighbour hopping along the two legs and between the two legs are t amd t’ respectively.

```
                    t
- - - ---o----o----o----o----o--  - -
         |    | t’ |    |    |
- - - ---o----o----o----o----o--  - -
```

Consider one orbital basis per site and derive the Hamiltonian in the said basis.
Plot the band-structure for a reasonable range of t/t’ to pin point the insulator to metal transition as we dicussed in class.

2. Add the diagonal hopping t’’ sequentially as shown below and show how this hopping inflences the transition. Keep |t’’| less than both |t| and |t’|.

(a)

```
- - - ---o----o----o----o----o--  - -
         | /  | /  | /  | /  |
- - - ---o----o----o----o----o--  - -
```

(b)

```
- - - ---o----o----o----o----o--  - -
         | X  | X  | X  | X  |
- - - ---o----o----o----o----o--  - -
```

3. (a) Consider square lattice done in class with nearest neighbour hopping t (= -2eV).
Plot the band-structure as a contour spanning the first BZand trace the Fermi curve
for 0.5e to 2e electron per unitcell on average in steps of 0.25e. You need to first sort all energies across the BZ to find the appropriate Fermi energies for the said average electron count.

(b) Add diagonal (next nearest neighbour) hopping in the square lattice and redo all asked in (a) for diagonal hopping elements -1eV and 1eV.